HHG/EUV Lab: Technical Repositioning Brief

Complete Findings, Sources, and Diagrams

Program context: Patent-and-lab portfolio supporting a Programmable Optical Transfer Function system. The `Laser-hhg-euv-lab` repository is being strengthened to serve as the generation-side wavelength-extension layer anchored by the `vdw-polaritonics-lab` (DUV: 193/248 nm intracavity modulation). This brief provides technically rigorous guidance on physics, complementarity, simulation scope, positioning language, and overclaim hygiene.

Executive Summary

Tabletop laser-driven high harmonic generation (HHG) is one of the most physically well-characterized routes to coherent extreme ultraviolet (EUV) and soft X-ray (SXR) radiation from compact laboratory systems. Its physics is grounded in the three-step model (ionization, electron acceleration, recombination) and described quantitatively by the Lewenstein strong-field approximation and macroscopic propagation models — the same family of models that underpin published, reproducible experimental results at major facilities including JILA, MPQ Garching, ELI-ALPS, and KMLabs.

The credible role for this repository is **not** to position HHG as a competing or replacement EUV lithography source (that is a solved, differently-scaled industrial problem owned by Sn-plasma LPP systems at ASML). The credible role is as the **generation-side and wavelength-range evidence layer** in a portfolio that claims programmable optical transfer function (OTF) concepts: HHG supplies the physical mechanism and wavelength-regime scaffolding by which the same control-architecture concepts that operate in the DUV (vdW cavity work) can be argued to extend toward shorter wavelengths.

The strongest complementarity between the two repos lies at the **front-end / driver-conditioning interface**: there is documented, experimentally supported evidence that temporal and spectral shaping of the driving laser (via AOPDF/Dazzler, chirped mirrors, adaptive wavefront correction) directly controls HHG spectral output, photon flux, and phase-matching efficiency. This is the interface where vdW-cavity-style control concepts are most honestly argued to connect.

The most important constraint to observe: **do not conflate the simulation of an architecture with experimental validation of a source**. All outputs from this lab should be labeled with explicit epistemic markers — "illustrative," "parameterized," or "architecture-level" — and the distinction between driver-side control (credible) and HHG medium / EUV beamline realization (engineering-constrained) must be maintained throughout.

Key Findings

A. HHG Physics and System Architecture

The three-step model and its quantitative consequences are well-established. The process — tunnel ionization, electron acceleration in the field, recombination with photon emission — gives rise to two analytically important scaling laws every serious reader will expect the repo to acknowledge:

Cutoff energy scaling:

$$E_{\text{cutoff}} = 3.17\,U_p + I_p \quad \text{where} \quad U_p \propto I\lambda^2$$

The ponderomotive energy U_p scales as the product of laser intensity I and the square of driver wavelength λ. Higher photon energies (shorter harmonic wavelengths, toward EUV) are accessible by increasing either I or λ. In practice, increasing I is ionization-limited (ground-state depletion destroys phase matching), so MIR drivers (1.8–3 μm) are used to push cutoffs toward 100+ eV (Physical Review Letters, Shiner et al. 2009; Attoworld Group, DESY).

Efficiency anti-scaling:

$$\eta(\lambda) \propto \lambda^{-(5 \text{ to } 6)}$$

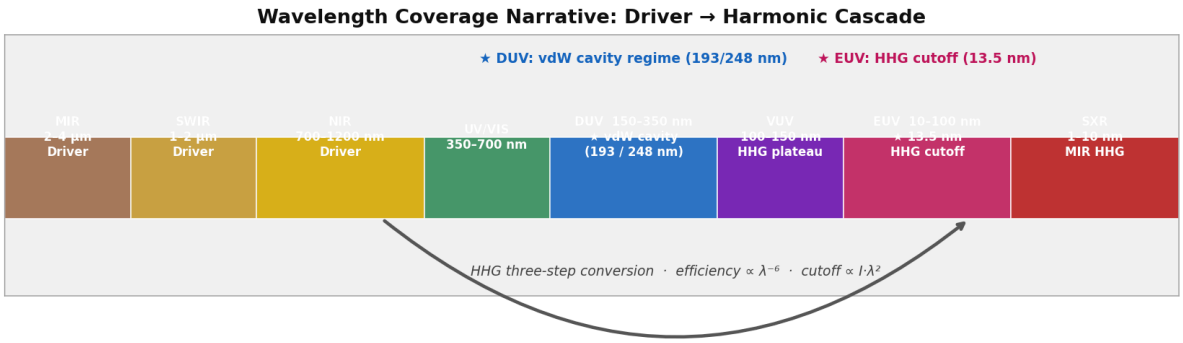
Single-atom HHG conversion efficiency falls dramatically with longer drivers — experimentally confirmed at $\lambda^{-6.3\pm1.1}$ in Xe and $\lambda^{-6.5\pm1.1}$ in Kr over 800–1850 nm (Shiner et al., PRL 2009). The efficiency and the cutoff energy are therefore in fundamental tension: the driver wavelength that reaches 92 eV (EUV at 13.5 nm) is the same one that suppresses conversion efficiency by several orders of magnitude relative to an 800 nm baseline. This is a physical constraint the repo must represent honestly — it is not a failure of engineering; it is a fundamental consequence of the three-step model.

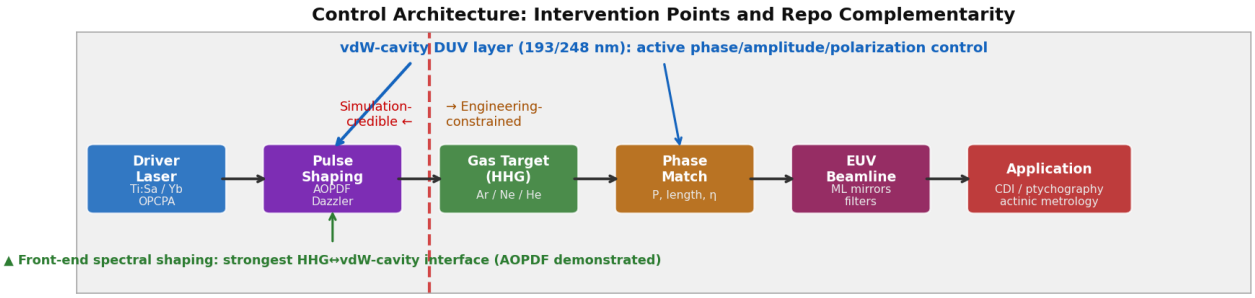
Phase matching is the macroscopic constraint governing usable photon flux. Perfect phase matching — where the coherence length L_{coh} exceeds the medium length — requires balancing neutral gas dispersion, plasma dispersion, and geometric (Gouy) phase. The classical phase-matching window exists only below a critical ionization fraction η_{crit}, typically ~3–9% depending on gas species and geometry (Nature Scientific Reports, Wikmark et al. 2022; HAL/CNRS review 2022). Phase matching effectively limits the harmonic orders that build coherently, and the pressure–medium-length hyperbolic tradeoff is an important modeling target for the repo.

Tabletop flux levels are well-characterized. Commercial systems (e.g., KMLabs XUUS 4 with Coherent Astrella) produce approximately 10^{11} photons/second/harmonic at 35 nm and 1.5×10^7 photons/second/harmonic at 13.5 nm (He gas) — with beamline throughput further reducing delivered flux to 0.1–10% of source (Coherent/KMLabs white paper). Do not assert or imply sources competitive with ASML's LPP (250 W at intermediate focus).

Pulse duration and yield interact non-monotonically. Below saturation, longer pulses yield more total XUV photons (linear dependence on pulse duration); at near-saturation

intensities, shorter pulses outperform because high free-electron density destroys phase matching in longer pulses (ELI-ALPS/ArXiv 2025). This creates a meaningful and physically defensible modeling variable.





B. Complementarity to vdW-Cavity Work

The vdW-cavity repo covers: DBR + hBN + graphene + MoS_2 + black phosphorus stack, intracavity phase/amplitude/polarization control, 193/248 nm DUV regime, and a programmable OTF concept applied at the emitter level. The HHG repo's genuine role is as the **generation chain** that extends the wavelength axis of the same OTF argument toward shorter wavelengths.

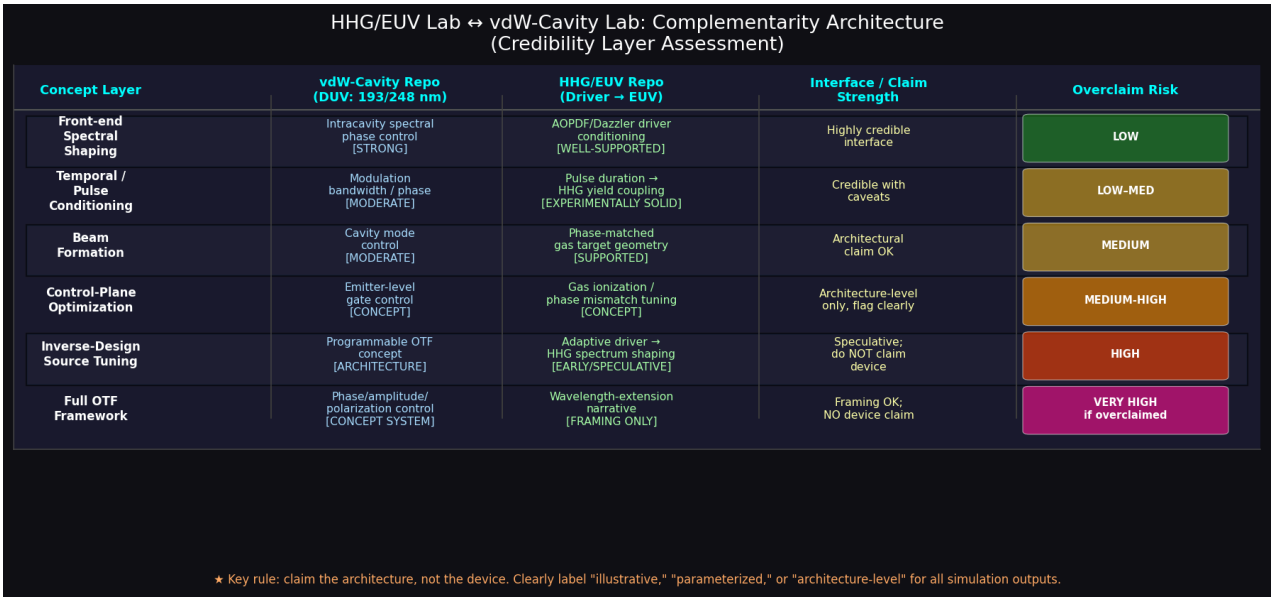
Six candidate interfaces exist, ranked by physical credibility:

Interface	Physical Mechanism	Credibility	Overclaim Risk
Front-end spectral shaping	AOPDF/Dazzler shapes driver spectrum/phase → directly controls HHG spectral output	STRONG — experimentally documented (Nature Sci. Reports, 2025)	Low
Temporal / pulse conditioning	Pulse duration → HHG yield / phase-matching tradeoff	WELL-SUPPORTED — multiple groups	Low
Beam formation / spatial mode	Structured beams (OAM, necklace, vector) → spatial and spectral HHG control	SUPPORTED — Science Advances 2022	Medium
Control-plane optimization	Adaptive feedback (wavefront sensor, Shack-Hartmann) → beam quality	DEMONSTRATED — Imagine Optic, MPQ	Medium

Interface	Physical Mechanism	Credibility	Overclaim Risk
Inverse-design source tuning	Optimization of driver waveform → target harmonic spectrum	EARLY — lab demos, not system-level	Medium-High
Full OTF framework across stages	Architecture argument extending vdW cavity OTF to HHG beamline	ARCHITECTURE-LEVEL ONLY	High if overclaimed

The **most defensible claim** is that the same programmable control plane that operates on the cavity-amplified DUV source can, at the driver-laser front end, apply spectral and temporal conditioning that systematically shifts the harmonic output of a connected HHG stage. AOPDFs (Dazzler) are standard components in HHG beamlines precisely because they provide this function (Rev. Mod. Phys. optimal control of HHG, 2008).

The **least defensible claim** is that the intracavity vdW modulation stack operates at EUV wavelengths. EUV (13.5 nm) is beyond the absorption edge of all vdW materials under consideration; MoS_2, graphene, hBN, and black phosphorus are not EUV-transparent. The cavity control architecture operates in DUV — the HHG stage is the EUV generation mechanism, downstream and physically separate.



C. Wavelength Coverage Narrative

The honest narrative is a **two-regime system with a defined interface**:

Regime 1 (vdW-cavity repo): 193–350 nm DUV. Programmable intracavity modulation. Active optical control architecture. Phase, amplitude, and polarization independently addressed per emitter layer.

Regime 2 (HHG repo): NIR/MIR driver (700–3000 nm) → nonlinear conversion → DUV/VUV/EUV/SXR harmonics (60–1 nm). Generation-side chain. Photon flux, coherence, and phase matching as the operative constraints.

The narrative bridge is: the same OTF control logic that applies to the cavity-level emitter stack can be extended, at the architectural level, to the driver-conditioning chain of an HHG source. This is the "programmable optical transfer function" claim applied to source generation rather than cavity modulation.

Wavelength regions the repo should model explicitly:

- **800 nm driver (Ti:Sa standard):** harmonics 17–51, covering ~30–55 nm (EUV/VUV plateau) — best-characterized, most validated simulation data
- **1030 nm driver (Yb:YAG, lower cost, increasingly common):** harmonics into 25–40 nm range
- **1800 nm driver (MIR):** lower efficiency but higher cutoff; access to 13–15 nm
- **Phase matching window** as a function of gas species (Ar, Ne, He, Kr), pressure, and focusing geometry
- **Beamline transmission:** filter stack (Al/Zr) and multilayer mirror throughput as a function of photon energy

D. Modeling Priorities

A credible HHG simulation lab should be structured as a hierarchy of model fidelity, each layer clearly labeled:

Layer 1 — Analytical / parameterized (fully credible for repo):

- Cutoff energy calculator: inputs are driver wavelength, peak intensity, gas species (I_p); output is maximum harmonic order and photon energy. Formula: $E_{\text{cut}} = 3.17U_p + I_p$, where $U_p[\text{eV}] = 9.33 \times I[10^{14}\ \text{W/cm}^2] \times \lambda[\mu\text{m}]^2$
- Efficiency scaling plot: relative single-atom yield as function of driver wavelength ($\propto \lambda^{-6}$)
- Phase matching pressure calculator: given gas species, focusing geometry (z_R), and ionization fraction, compute coherence length L_{coh} and compare to medium length
- Beamline loss calculator: compound transmission through N mirrors at reflectivity R each, M filter layers at transmission T each

Layer 2 — Semi-classical / SFA-level (credible for architectural simulation):

- Lewenstein model single-atom dipole response: harmonic spectrum shape, cutoff, plateau structure. Open-source implementation available (github.com/mightymightys/Lewenstein-HHG)
- Short vs. long trajectory separation (affects spatial coherence and phase-matching window)
- Saddle-point approximation for electron trajectories

Layer 3 — Macroscopic propagation (architecture-level claims only):

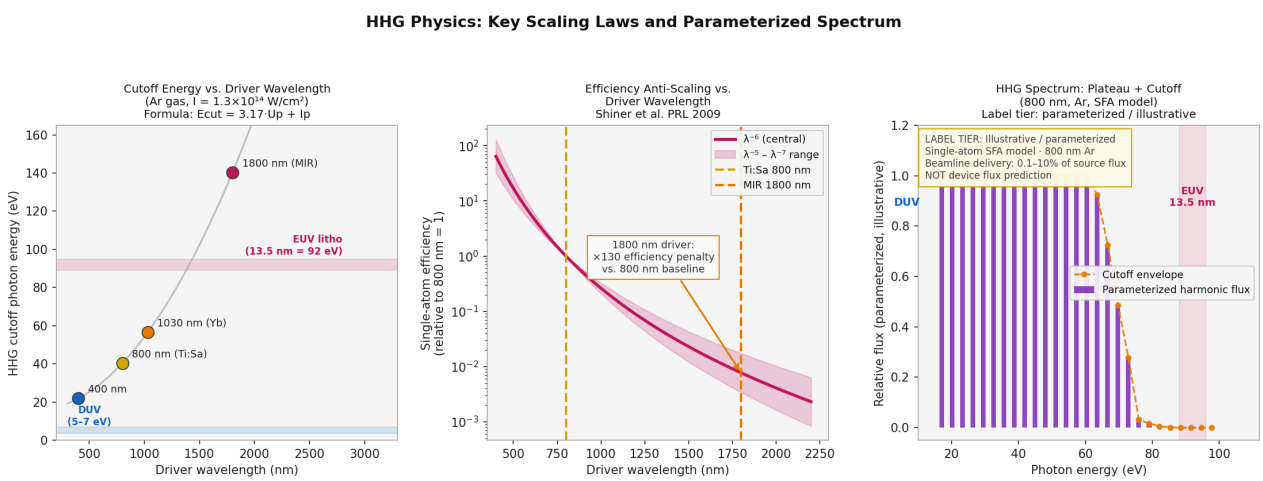
- 3D non-adiabatic propagation (wave equation + Lewenstein polarization source term): computationally expensive, appropriate only if the repo targets quantitative flux prediction
- Open-source C++ program published September 2025, BSD-licensed, handles both macroscopic and microscopic aspects (ArXiv 2509.02867)
- Multiscale TDSE/SAE approach for beyond-SFA accuracy (Physical Review A, 2023)

User-tunable parameters the repo should expose:

- Driver wavelength (400–3000 nm continuous)
- Peak intensity (0.5–5 × 10^{14} W/cm²)
- Gas species (Ar, Ne, He, Kr, Xe) with corresponding I_p values
- Gas pressure (1–150 mbar) and medium length (1–30 mm)
- Focusing geometry: beam waist and Rayleigh length
- Pulse duration (5–100 fs)

Outputs explicitly labeled by epistemic tier:

- "Analytical / formula-based": cutoff, efficiency scaling
- "Parameterized model / illustrative": SFA spectrum shape
- "Architecture-level": OTF framing, control-chain diagram
- Never label anything "simulated device performance" without macroscopic 3D propagation validation



E. Patent and Positioning Hygiene

Language That Overclaims

Overclaiming Language	Why It Fails	Defensible Replacement
"EUV source based on vdW cavity modulation"	vdW materials absorb at EUV; cavity does not operate at 13.5 nm	"DUV cavity modulation architecture feeding an HHG generation chain"

Overclaiming Language	Why It Fails	Defensible Replacement
"High-brightness tabletop EUV for lithography"	Tabletop HHG is 8+ orders of magnitude below LPP in power	"Coherent tabletop EUV suitable for metrology, CDI, and actinic inspection"
"Programmable EUV source"	HHG tunability is limited; full programmability is an architecture concept	"Programmable driver conditioning architecture for HHG source control"
"Demonstrates EUV generation via vdW-enhanced cavity"	No demonstrated HHG from vdW-cavity stack	"Models the control architecture for a system in which a vdW-cavity DUV stage conditions the driver of an HHG conversion chain"
"Inverse-designed EUV source"	System-level inverse design of HHG not demonstrated	"Architecture study applying inverse-design principles to driver-conditioning for HHG optimization"
"Simulated EUV output consistent with device performance"	Conflates SFA single-atom model with macroscopic device flux	"Parameterized analytical model illustrating cutoff energy and efficiency trends"

The Four Epistemic Tiers

Every output in the repo should be labeled according to one of four tiers:

1. **Demonstrated subsystem behavior:** something physically built and measured (e.g., AOPDF pulse shaping of a Ti:Sa driver is demonstrated). This tier is the province of cited references, not this repo's own claims.
2. **Experimentally feasible architecture:** a plausible combination of demonstrated subsystems with a credible physical path to integration. Claim: "consistent with published experimental feasibility."
3. **Simulated / parameterized model:** analytical or SFA-level calculation, correctly labeled, with stated assumptions and known limits of model validity.
4. **Architecture-level framing:** the OTF narrative, system diagram, control-plane concept. Claim nothing about device performance.

EPO/USPTO Simulation Patent Hygiene

The EPO Enlarged Board G1/19 (2021) established that computer-implemented simulations are patentable in principle but require that the **intended technical use producing a technical effect be at least implicitly specified in the claims** (EPO G1/19, Mathys & Squire analysis). For USPTO, the Alice framework requires that a claim directed to a simulation include an inventive concept sufficient to move beyond an abstract idea — tying upstream hardware architecture features to downstream algorithm steps is the recommended approach for optical device claims (SPIE Photonics Focus, 2021). The practical implication: claims should specify the control data produced by the simulation as inputs to a physical device (the driver laser conditioning system), not as standalone simulation results. EPO guidelines (EPO 3.3.2) require that for purely numerical simulations, the intended technical use must be explicitly or implicitly specified in the claim scope.

Strongly Supportable Claims

The following can be stated with confidence in repo documentation, a website, or a patent specification (as "experimentally feasible" or "architecture-level"):

1. Three-step model + SFA-level harmonic spectrum.
Cutoff energy, plateau structure, and efficiency scaling as functions of driver wavelength, intensity, and gas species are quantitatively predictable. Model outputs can be labeled "parameterized analytical model" without qualification.

2. Driver front-end conditioning controls HHG spectral output.
AOPDFs (Dazzler) and chirped mirrors are commercially deployed for exactly this purpose in active HHG beamlines. Claiming that the same control principles that apply to vdW-cavity spectral and phase management can be architecturally extended to driver conditioning is physically honest and documented (Nature Sci. Reports OPCPA/Dazzler, 2025; Amplitude Laser Dazzler).

3. Phase matching windows are tunable by gas pressure and medium length.
Demonstrated experimentally including by pre-ionization methods that extend the phase-matched bandwidth by pre-selecting ionization fraction before the driving pulse arrives (Nature Scientific Reports 2022).

4. Tabletop HHG is the credible source for EUV actinic metrology, mask inspection, CDI, and ptychography.
Multiple groups have demonstrated this: KMLabs CDI at 22 nm resolution (KMLabs); HHG ptychography at wavelength-scale resolution (Scientific Reports 2019); actinic EUV patterned mask inspection (Semantic Scholar 2023); HHG-CDI for semiconductor metrology (VU Amsterdam 2025). These are the correct use-case frame.

5. Cavity-enhanced HHG (CE-HHG) is a demonstrated architecture enabling MHz-rate EUV at 10–100 eV, first demonstrated 2005 (JILA/MPQ), now mature enough for commercial-grade photoemission spectroscopy (ArXiv 2410.11589, 2024). CE-HHG is the most credible precedent for cavities playing a control role in HHG chains — this is the strongest interface point with the vdW-cavity framing, provided CE-HHG and vdW cavities are kept physically distinct in the language.

6. Spatial mode control of the driver transfers to the harmonic spatial profile.
OAM, necklace beam, and vector beam driving demonstrated experimentally to control HHG spectral spacing, divergence, and mode structure (Science Advances 2022; PRL 2014). Beam formation claims are architecturally supportable.

7. MIR-driven HHG reaches 13.5 nm.
Demonstrated with He gas and millijoule-class OPCPA systems (Attoworld/DESY). Accurately represent that 13.5 nm is physically accessible but with flux 4–5 orders of magnitude below Ar-driven harmonics at 35 nm.

Use Caution / Do Not Overclaim

1. Do not claim industrial EUV source relevance.
ASML's LPP source delivers 250 W at 13.5 nm intermediate focus. Tabletop HHG delivers nanowatts to microwatts. The gap is not engineering; it is physics (conversion efficiency, duty cycle, repetition rate, plasma vs. gas target). Any statement about "scalable to production" or "alternative to LPP" will destroy credibility with expert readers.

2. Do not claim vdW materials operate at EUV wavelengths.
MoS_2, graphene, hBN, and black phosphorus all have strong absorption below ~100 nm. The intracavity modulation architecture operates at DUV (193/248 nm), not EUV. The HHG stage is downstream and separate.

3. Do not conflate CE-HHG cavities with the vdW cavity.
CE-HHG cavities are femtosecond enhancement cavities operating at NIR driver wavelength, passively amplifying pulses by coherent addition. The vdW cavity is an active modulation structure at DUV. These are physically distinct systems. Framing them as the same "cavity concept" is misleading to expert readers.

4. Do not claim conversion efficiency improvement without quantitative basis.
Any statement that driver conditioning "improves HHG efficiency" must be referenced to specific demonstrated efficiency gains or labeled "architectural hypothesis, not validated."

5. Do not present SFA/Lewenstein model outputs as device flux predictions.
The Lewenstein model computes single-atom response; macroscopic flux requires 3D propagation with plasma, absorption, and focusing geometry included. Labeling SFA output as "source brightness" is a significant overclaim.

6. Do not present the wavelength coverage narrative as implying a single integrated device.
The DUV cavity and the HHG chain are architecturally complementary but are not demonstrated as a single integrated system. Frame as a portfolio argument, not a device claim.

7. Avoid "inverse-designed EUV source" without qualification.
Inverse design of a driver waveform to produce a target harmonic spectrum has been demonstrated at the laboratory level for specific cases, but system-level deployment is an early-stage research direction. The term implies a complete optimizable system, which is not currently the state of the field.

Recommended Repo Improvement Priorities

Priority 1 — Epistemic clarity (immediate, highest leverage):
Add a README section titled "Evidence Architecture and Claim Taxonomy." Define the four tiers and tag every output. This is the single highest-credibility signal to an expert reader — it demonstrates rigor before they read a single equation.

Priority 2 — Analytical models (high value, zero overclaim risk):
Implement and display as properly labeled calculators:

- Cutoff energy as a function of driver wavelength and intensity (formula-based, exact)
- Efficiency scaling chart (λ^{-5} to λ^{-7} band) vs. driver wavelength
- Phase matching window calculator (pressure, gas species, medium length)
- Beamline transmission calculator (filter + mirror stack)

Priority 3 — Wavelength architecture diagram (high strategic value):
Build an interactive wavelength axis from 3 μm to 1 nm annotating: driver wavelength band, harmonic order range, approximate phase-matched flux, industrially relevant anchor wavelengths (13.5 nm EUV litho, 30 nm actinic inspection, 193/248 nm DUV), and the vdW-cavity operating point. This single figure communicates the complementarity portfolio argument immediately to any expert reader.

Priority 4 — SFA-level simulation (medium priority):
Integrate a Lewenstein-model single-atom response calculator for Ar at 800 nm and Ne at 1030 nm. Label clearly as "single-atom parameterized model." Reference the open-source C++ HHG simulation program (GitLab caschroeder/hhg) and the GitHub Lewenstein implementation (github.com/mightymightys/Lewenstein-HHG).

Priority 5 — Control architecture narrative (core patent positioning):
Add a driver-conditioning module that shows how AOPDF spectral phase and amplitude shaping maps to changes in HHG output (harmonic order weighting, pulse duration, spectral width). This is the interface between the vdW-cavity OTF concept and the HHG generation chain — the core novel claim architecture.

Priority 6 — Actinic/metrology use-case section (low risk, high framing value):
Document: CDI, ptychography, actinic EUV mask inspection, ARPES with CE-HHG. Positions the repo as relevant to semiconductor metrology without overreaching into lithography source territory.

Priority 7 — Explicit LPP disclaimer:
Add a "Scope and Limitations" section stating directly: "This architecture is relevant to research, metrology, and inspection applications where coherence and tunability are primary requirements. High-volume manufacturing EUV lithography uses laser-produced plasma (LPP) sources at power levels not accessible to gas-jet HHG." This pre-empts the most common expert objection and signals intellectual honesty.

Recommended Figures, Models, and Interactive Controls

Output	Type	Label Tier	Technical Basis
Wavelength coverage map (driver → harmonic)	Static / interactive diagram	Architecture-level	Three-step model
Cutoff energy vs. driver λ and I	Interactive calculator	Parameterized analytical	Exact formula
Single-atom efficiency vs. driver λ	Chart with uncertainty band	Parameterized analytical	Shiner et al. PRL 2009

Output	Type	Label Tier	Technical Basis
Phase matching window (P, L, gas)	Interactive calculator	Parameterized analytical	Classical PM theory, HAL/CNRS 2022
HHG harmonic spectrum (SFA)	Parameterized model output	Parameterized model (SFA)	Lewenstein 1994
Beamline transmission calculator	Table / calculator	Parameterized analytical	Published optics reflectivity data
AOPDF → HHG control architecture diagram	System diagram	Architecture-level	CE-HHG and AOPDF literature
vdW cavity ↔ HHG chain interface diagram	Complementarity map	Architecture-level	Both repos
Use-case matrix (CDI, ptychography, metrology, ARPES)	Reference table	Architecture-level	Published HHG applications

Suggested Positioning Language

For the Repo README

> **Laser-HHG-EUV Lab** is a simulation and architecture study repository supporting a broader programmable optical systems portfolio. It models the generation-side chain of a laser-driven high harmonic generation (HHG) system, providing analytical calculators, parameterized spectral models, and architectural diagrams that extend the wavelength narrative of the companion `vdw-polaritonics-lab` from the DUV (193/248 nm) intracavity modulation regime toward shorter wavelengths in the EUV and soft X-ray bands.
> All outputs are labeled according to their epistemic tier: (1) formula-based analytical models, (2) parameterized simulation models (single-atom SFA level), or (3) architecture-level system diagrams. No output represents a demonstrated device. The physical constraints of tabletop HHG — conversion efficiency scaling, phase-matching limits, and beamline throughput — are represented explicitly and honestly. This repository is relevant to research, metrology, and actinic inspection applications where coherence and tunability are primary requirements.

For the Microsite

> The HHG/EUV Lab extends the programmable optical transfer function (OTF) architecture into the generation domain. Where the vdW-cavity system applies active phase, amplitude, and polarization control at 193–248 nm, the HHG chain provides the mechanism by which the same control architecture — applied at the driver-laser front end — translates into structured harmonic output across DUV, VUV, and EUV wavelength bands. The interface between these two systems is the driver-conditioning stage: spectral and temporal shaping of the near-infrared or mid-infrared driving pulse directly governs harmonic order weighting, phase-matched bandwidth, and coherence properties of the EUV output.

For a Patent Specification

> The system described herein comprises a first optical control stage operating in the deep ultraviolet spectral region (DUV: 150–350 nm), wherein an intracavity modulation structure comprising layered van der Waals materials applies independently programmable phase, amplitude, and polarization transformations to an optical field; and a second stage comprising a driver laser conditioning subsystem and a gas-medium high harmonic generation cell, wherein the driver conditioning subsystem applies spectral phase and amplitude programming to a near-infrared or mid-infrared driving pulse such that the spectral and temporal properties of the high harmonic output are modified in accordance with a target optical transfer function defined at the system level. The driver conditioning subsystem implements control actions that are governed by the same programmable transfer function architecture as the intracavity modulation stage, thereby extending the OTF control plane from the DUV operating regime of the first stage through the harmonic conversion process of the second stage toward extreme ultraviolet photon energies.
> *[Note: enumerate subsystems with "configured to" language per MPEP §2114; label as architecture claim, not device performance claim; do not assert demonstrated flux or brightness figures for the HHG stage.]*

Reference List

Reference	Quality / Notes
Shiner et al., PRL 103, 073902 (2009)	Foundational — definitive experimental λ^{-6} efficiency scaling measurement, 800–1850 nm in Xe and Kr
Lewenstein et al., Phys. Rev. A 49, 2117 (1994)	Foundational — the strong-field approximation (SFA) model underlying all analytical HHG simulation
ELI-ALPS / ArXiv 2507.10153 (2025)	Recent primary — systematic study of HHG yield vs. pulse duration; ELI-ALPS SYLOS laser system
Wikmark et al., Nature Scientific Reports (2022)	Strong — phase matching with pre-ionized medium; analytical model + experiment; capillary discharge
HAL/CNRS optimization review (2022)	Strong — comprehensive analytical treatment of pressure–length hyperbolic optimization for HHG CE
Carstens et al. / ArXiv 2410.11589 (2024)	Strong review — cavity-enhanced HHG at >10 MHz; state of the field for CE-HHG photoemission
Coherent/KMLabs XUUS 4 white paper	Industrial benchmark — measured flux at 35 nm (10^{11} ph/s) and 13.5 nm (1.5×10^{7} ph/s)
KMLabs CDI 22 nm resolution	Strong application — best demonstrated CDI resolution from tabletop HHG (13 nm source)
Scientific Reports ptychography (2019)	Strong — wavelength-scale HHG ptychography with record photon flux; 2.5λ resolution
Science Advances necklace-driven HHG (2022)	Strong — OAM/necklace beam driver controls HHG spectral spacing and divergence

Reference	Quality / Notes
Physical Review X phase matching (2020)	Strong — full HHG dipole phase-mismatch treatment; long-wavelength few-cycle drivers
PRL cavity-enhanced HHG + spatial modes (2014)	Strong — HHG in 78 MHz enhancement cavity with non-Gaussian transverse modes
Rev. Mod. Phys. optimal control of HHG (2008)	Review, authoritative — adaptive pulse shaping for HHG spectral control; hollow-core fiber
Nature Scientific Reports OPCPA for HHG (2025)	Recent — dual SWIR/MIR OPCPA with Dazzler control → direct HHG spectral tuning demonstration
Actinic EUV mask inspection (Semantic Scholar 2023)	Industry application — HHG actinic patterned mask inspection; Lasertec ACTIS context
VU Amsterdam HHG-CDI thesis (2025)	Strong review — tabletop HHG-CDI for semiconductor metrology; ptychography + holography
ArXiv 2509.02867 open-source HHG simulation (2025)	Recent, practical — BSD-licensed C++ HHG simulation program, macroscopic + microscopic
Phys. Rev. A multiscale HHG simulation (2023)	Strong — TDSE/SAE multiscale macroscopic simulation; Ar at two scales
EPO G1/19 simulation patentability (2021)	Legal, authoritative — Enlarged Board decision on computer simulation patent eligibility
EPO Guidelines 3.3.2 — simulation, design, modelling	Legal, authoritative — EPO examination practice for simulation claims under G1/19
SPIE patenting optical inventions (2021)	Practical legal guidance — Alice risk mitigation for optical device claims; hardware-algorithm tie
Attoworld/DESY HHG cutoff scaling	Strong — cutoff energy confirmation and laser intensity calibration with 1800 nm MIR driver
Imagine Optic HHG wavefront control	Industrial — EUV wavefront sensors and adaptive optics for HHG beamline optimization
GitHub Lewenstein-HHG (mightymightys)	Practical — GPL-licensed Python SFA simulation; single-atom level, arbitrary driving waveforms
GitLab caschroeder/hhg	Practical — BSD C++ HHG simulation; macroscopic propagation + Lewenstein microscopic response

Brief prepared April 2026. All simulation claims in the repository should be labeled according to the epistemic tier taxonomy defined in Section D. No content in this brief constitutes a patent opinion. Consult qualified patent counsel before filing claim language derived from this document.